$$|S_p - S_{m^2}| \le M^2 \sum_{j=m^2+1}^{p} \frac{1}{j^2} \le M^2 \sum_{j=m^2+1}^{p} \frac{1}{j^2} \le M^2 \frac{p-m^2}{(m^2+1)^2} \to 0\,.$$

所以 $|f(x) - S_p(x)| \le |f(x) - S_{m^2}(x)| + |S_{m^2}(x) - S_p(x)| \to 0$, a.e..

29. 考虑函数 $\chi_A = \sum_{k=1}^{\infty}(\chi_A, \varphi_k)\varphi_k$ 的范数，利用 Schwarz 不等式证明.

30. 利用第 29 题的结论反证.

§4.3

4. 利用函数 f 的积分平均连续性证明.

5. 令 $s, l \in [0,1]$ 待定，令 λ, λ' 和 σ, σ' 分别为待定共轭指数. 直接计算:

$$\begin{aligned}
|f*g(x)| &\le \int_{\mathbf{R}} |f(x-y)|\,|g(y)|^l |g(y)|^{1-l}\mathrm{d}y \\
&\le \left(\int_{\mathbf{R}} |f(x-y)|^{\lambda}|g(y)|^{l\lambda}\mathrm{d}y\right)^{1/\lambda}\left(\int_{\mathbf{R}} |g(y)|^{(1-l)\lambda'}\mathrm{d}y\right)^{1/\lambda'} \\
&= \left(\int_{\mathbf{R}} |f(x-y)|^{s\lambda}|f(x-y)|^{(1-s)\lambda}|g(y)|^{l\lambda}\mathrm{d}y\right)^{1/\lambda}\left(\int_{\mathbf{R}} |g|^{(1-l)\lambda'}\mathrm{d}y\right)^{1/\lambda'} \\
&\le \left(\int_{\mathbf{R}} |f(x-y)|^{s\lambda\sigma}\mathrm{d}y\right)^{1/\sigma\lambda}\left(\int_{\mathbf{R}} |f(x-y)|^{(1-s)\lambda\sigma'}|g(y)|^{l\lambda\sigma'}\mathrm{d}y\right)^{1/\sigma'\lambda} \\
&\quad \cdot \left(\int_{\mathbf{R}} |g|^{(1-l)\lambda'}\mathrm{d}y\right)^{1/\lambda'}.
\end{aligned}$$

设 $\begin{cases} p = s\lambda\sigma, \\ q = (1-l)\lambda', \\ r = \sigma'\lambda, \\ p = (1-s)\lambda\sigma'. \end{cases}$ 解得

$$\lambda = p, \quad \sigma\lambda = rp/(r-p), \quad l\lambda\sigma' = q, \quad 1/\lambda' = 1/q - 1/r.$$

所以

$$|f*g(x)| \le \|f\|_p^{(r-p)/r}\left(\int_{\mathbf{R}} |f(x-y)|^p|g(y)|^q\mathrm{d}y\right)^{1/r}\left(\int_{\mathbf{R}} |g|^q\mathrm{d}y\right)^{1/q-1/r},$$